SVKM's NMIMS | Mukesh Patel School of Technology Management & Engineering

Machine Learning | B.Tech Semester V | AY 2026-27

DemandIQ

Machine learning for product demand and stockout risk prediction

Forecast what will sell • warn what will run out • say how much to order • explain why

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Guide: Prof. _____

Every store asks one question: how much should we stock?

↓ Too little stock

- The shelf runs empty - the sale is lost
- The customer often goes to a competitor
- Empty shelves damage trust in the store

↑ Too much stock

- Cash is locked up in goods
- Storage and handling costs keep adding up
- Old stock is discounted or thrown away

8.3%

of items are out of stock on an average day in retail worldwide

Gruen, Corsten & Bharadwaj (2002)

Root cause: nobody knows tomorrow's demand. Rules like "order last month's sales plus a bit" ignore promotions, holidays and weekdays - so stores run short exactly when demand jumps.

OBJECTIVES

Four objectives, each building on the last

1

Predict demand

Daily sales per store for the next 7 days, with a P10-P90 range

e.g. store 262, Mon 1 Jun: 24,078

2

Predict stockout risk

Probability of running out in the next 3 days

e.g. store 122: 30% → MEDIUM

3

Recommend the order

Order-up-to level from forecast + uncertainty; emergency order if risk is high

e.g. order up to 26,856

4

Explain + what-if

SHAP reasons for every number; test promotion, holiday, competitor, price

e.g. promotion: +29% to +45%

Supporting objective: group the 935 stores into segments with K-Means so one plan does not have to fit all.

What 20 books and 15 research papers told us

20

reference books
forecasting, inventory theory,
ML, explainability, causality

15

research papers
mostly 2019 - 2026,
IJF, EJOR, NeurIPS, IEEE Access

1. **ML beats statistics on retail data**
   M5 competition: LightGBM-based methods won on Walmart data *Makridakis et al., 2022*
2. **Promotions and special days need explicit features**
   Regression trees with promo flags; calendar special days *Gür Ali 2009; Huber 2020*
3. **Stockout prediction is rare - and needs stock + forecast**
   1.6 M SKUs: stock level and near-term forecast matter most *Liu et al., 2025*
4. **A more accurate forecast is not always a better decision**
   Weak link between accuracy and inventory cost on M5 *Theodorou et al., 2025*
5. **Explanations stop at the forecast**
   LightGBM + SHAP/ICE explain demand, not risk or orders *Mishra et al., 2026*

Six gaps - and how DemandIQ answers each one

G1 **Forecast, risk and ordering studied separately**

→ One pipeline: forecast → risk → order

G2 **Explanations end at the forecast**

→ SHAP for the demand AND the risk model

G3 **Accuracy is not decision quality**

→ Judge on stockouts, fill rate, stock and cost

G4 **What-if read as cause and effect**

→ Every what-if labelled as model behaviour

G5 **One policy for very different stores**

→ K-Means segments; uplift per segment

G6 **No public data has stock on hand**

→ Calibrated inventory replay over real sales

DATASET

Real sales from 1,115 Rossmann drugstores, 2013 - 2015

1.02 M	942	935	27
daily store records	days of history	stores with complete history	leak-free features

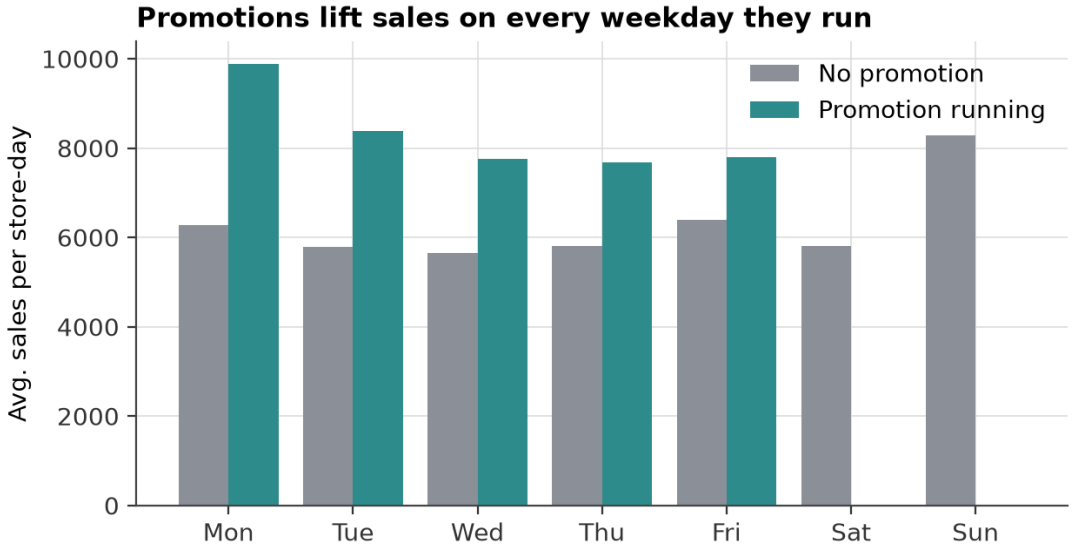
Split strictly by date - never shuffled

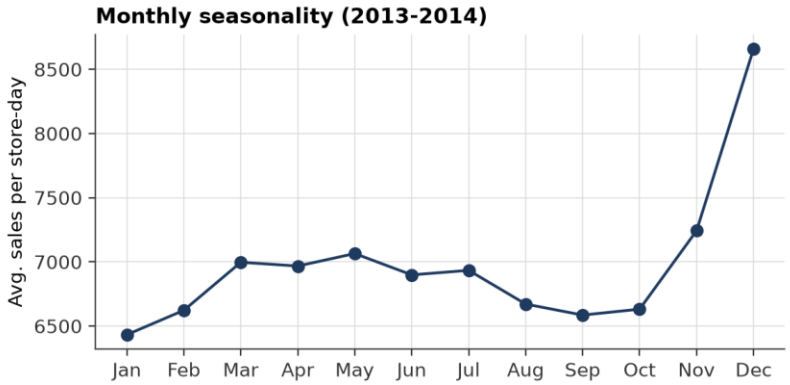
Training	Validation	Test
Jan 2013 - Mar 2015	Apr - May	Jun - Jul 2015

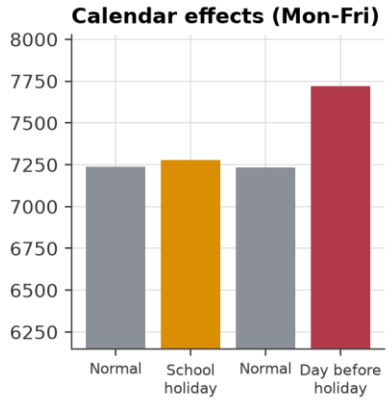
Why Rossmann? Real retail data with exactly the drivers we need - promotions, public and school holidays, weekday, season, competition and store format. Public, free and a well-known benchmark.

Cleaning: dropped 180 stores with a 6-month refurbishment gap, re-indexed every store to a full calendar, every sales feature is at least 7 days old.

Promotions add 40% - and timing matters







Promotions

run on 45% of trading days and lift sales by about 40%

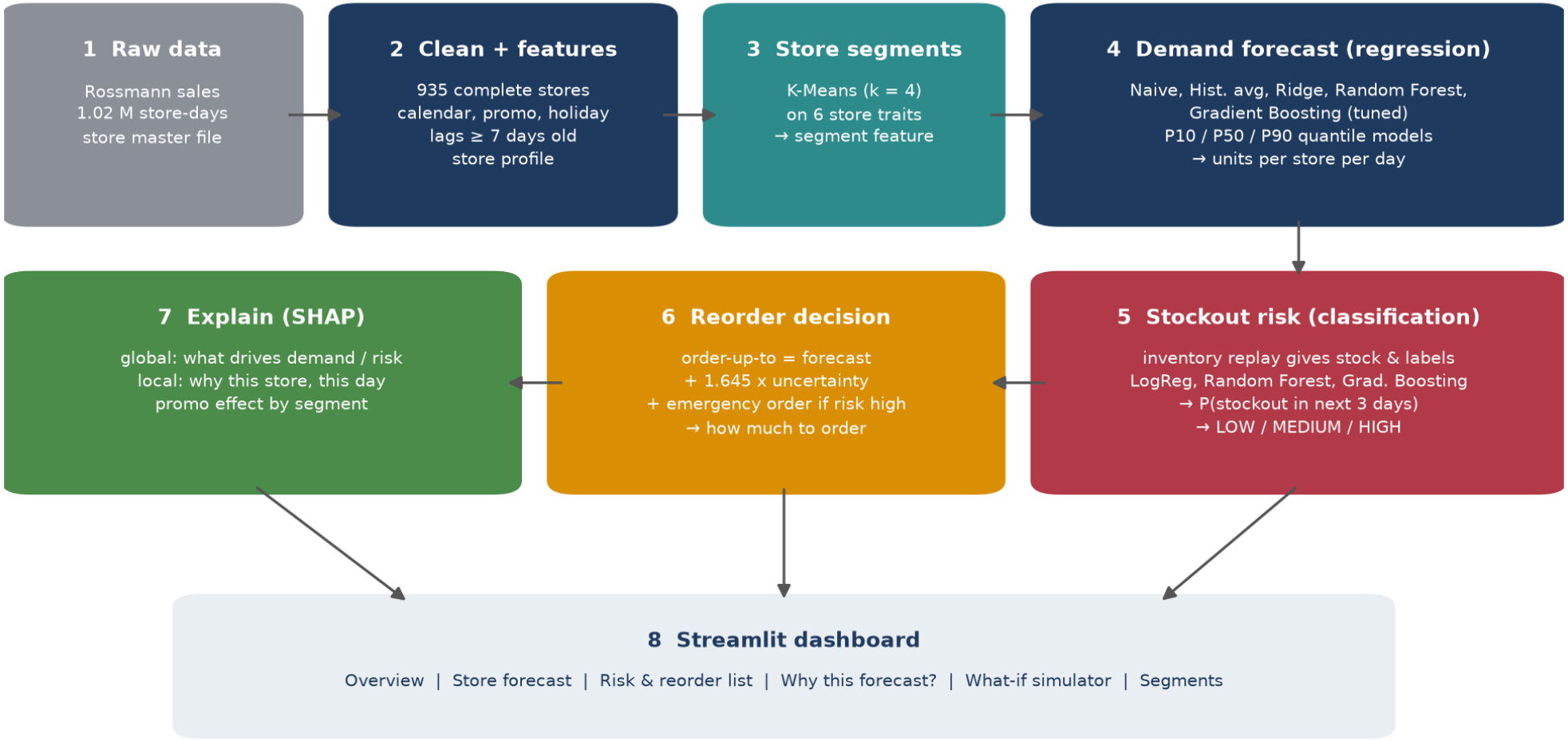
Weekly rhythm

Monday is the busiest day; most stores close on Sunday

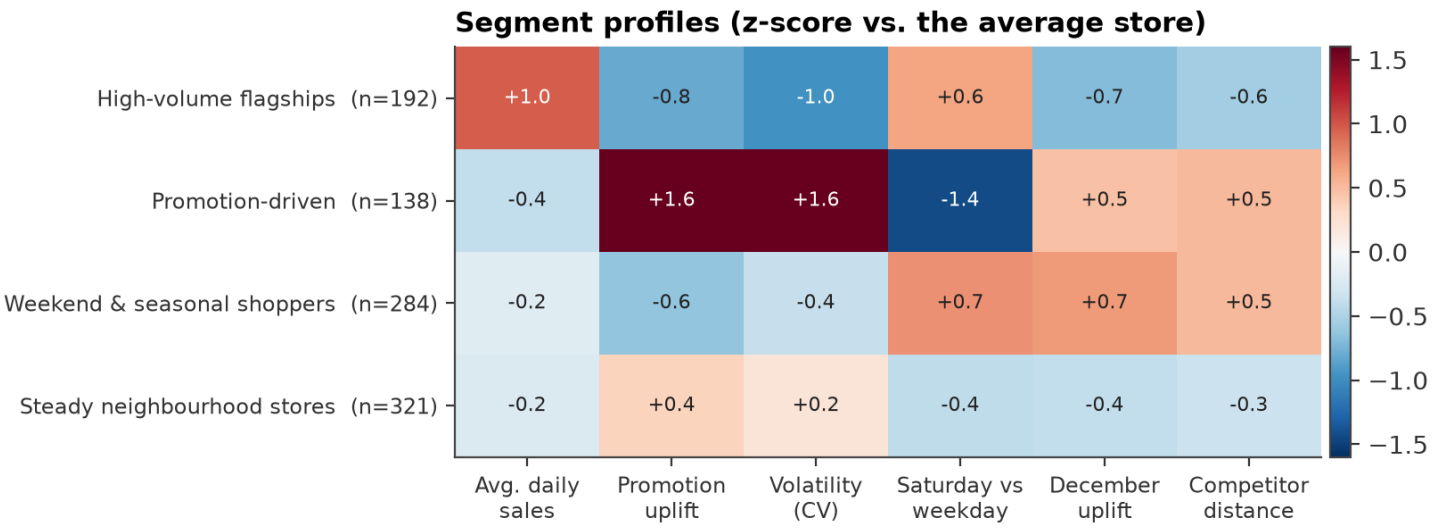
Calendar

December peak; extra sales in school holidays and before public holidays

Eight steps from raw sales to an explained order



K-Means finds four kinds of stores



1. **High-volume flagships (192)**
   9,292/day, steady, close competitors

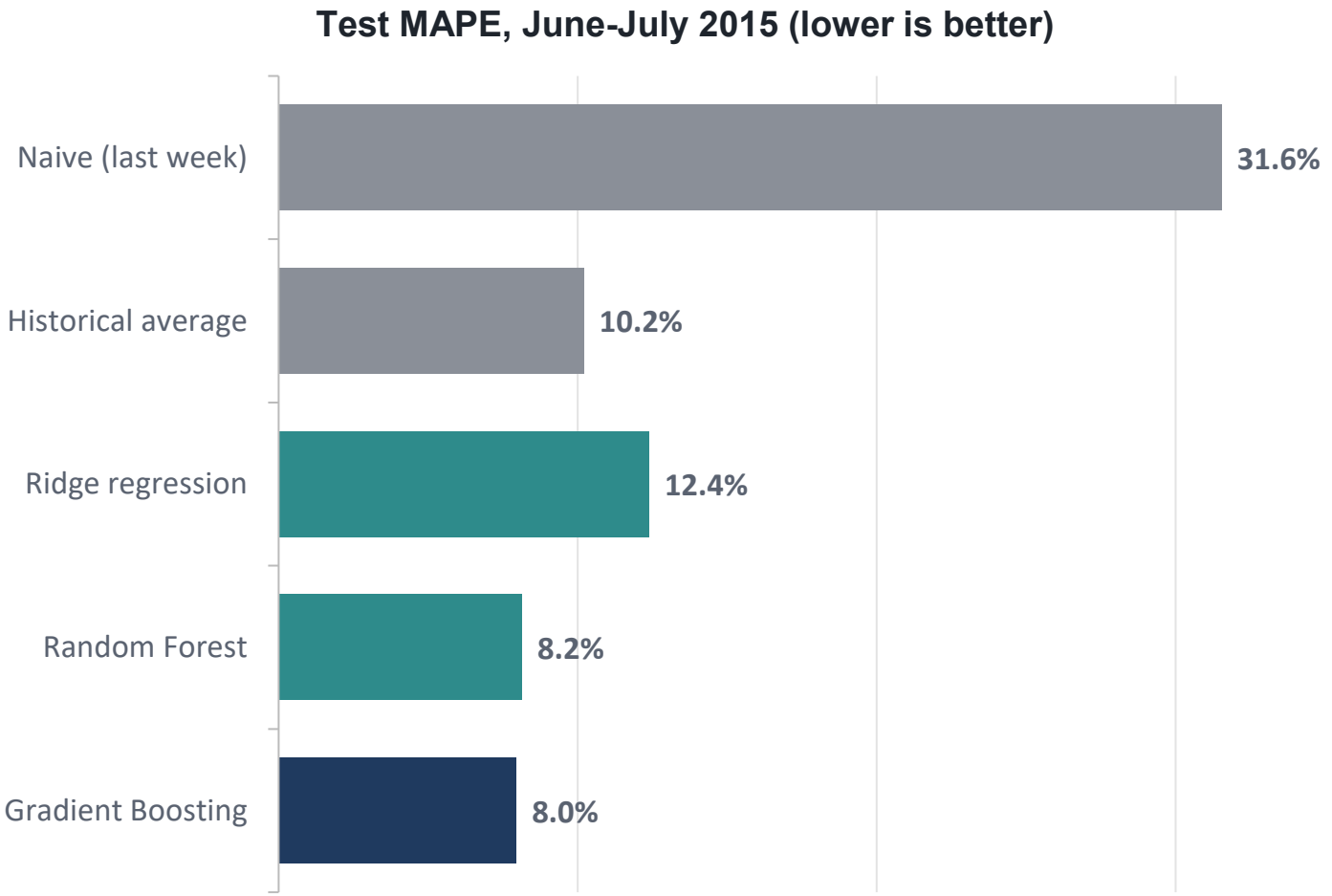
2. **Promotion-driven (138)**
   +72% on promotion days, most volatile

3. **Weekend & seasonal (284)**
   strong Saturdays and December

4. **Steady neighbourhood (321)**
   close to average on every trait

Why it matters: the same promotion lifts a promotion-driven store far more than a flagship, so one promotion plan and one safety buffer do not fit all stores. The segment is also a feature for both models.

Gradient Boosting cuts forecast error to 8.0%



0.941

R^2 on the test window

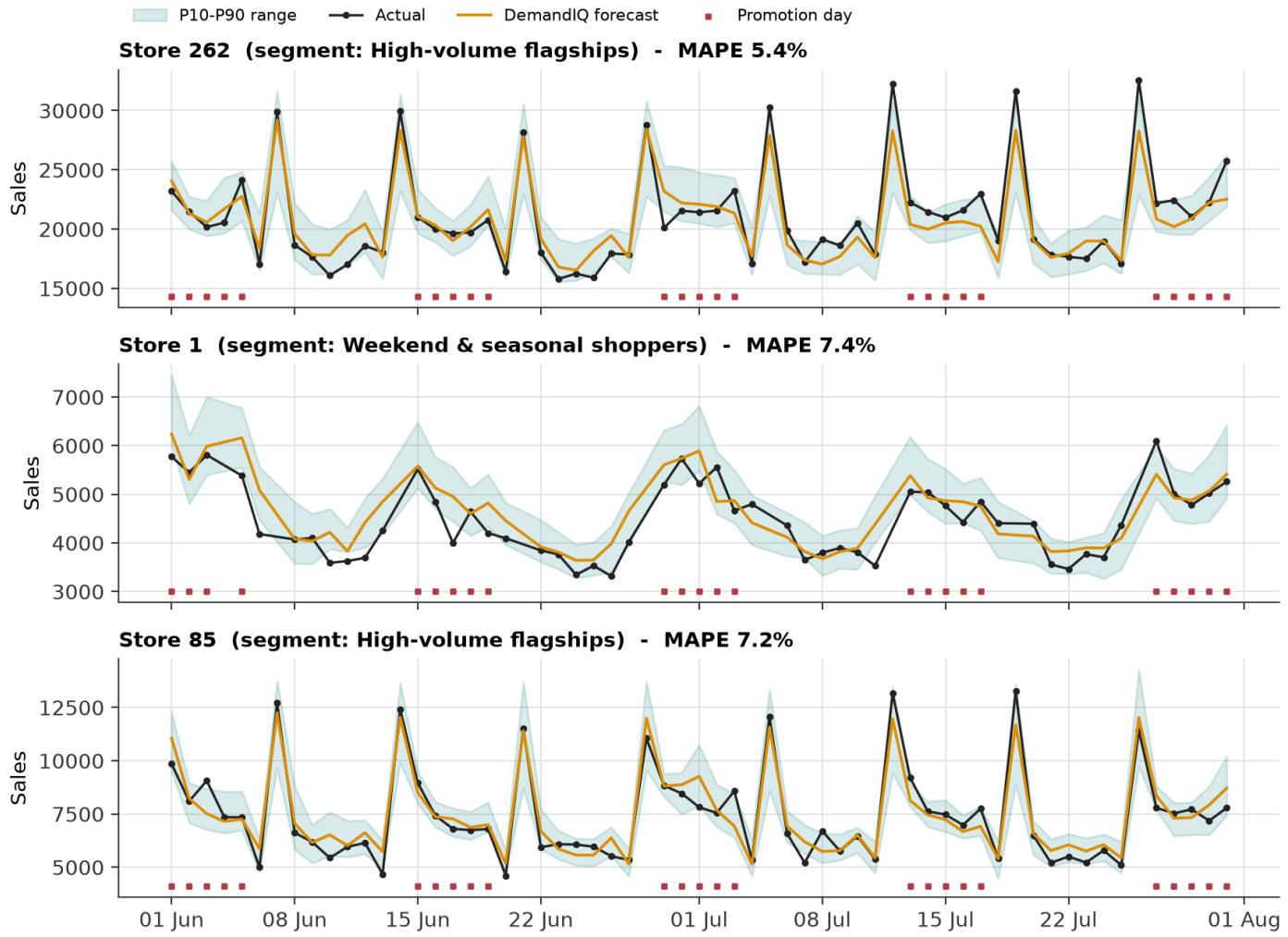
+0.4%

bias - almost no systematic over- or under-forecast

73%

of days inside the P10-P90 range (target 80%)

The forecast follows promotions and Monday peaks



Leak-free by design

Every sales feature is at least 7 days old, so one model forecasts the whole coming week.

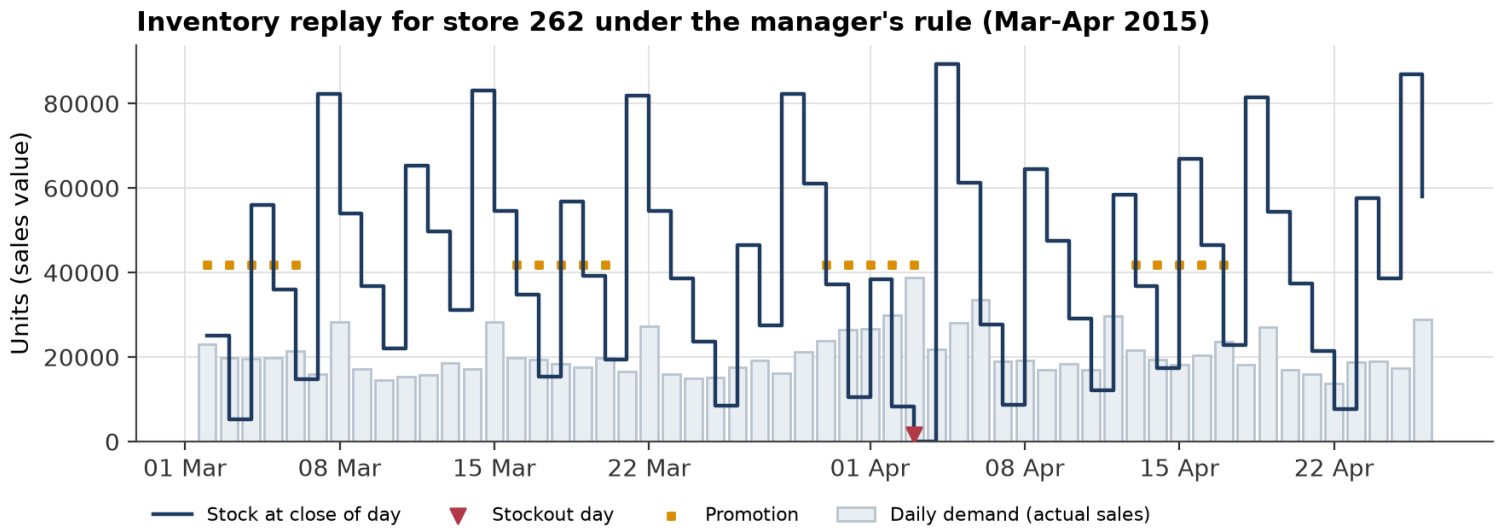
Uncertainty, not just a number

Quantile models give P10 and P90; the band feeds the safety stock.

Honest tuning

Grid search on Apr-May; the test window was opened only once.

Rossmann has no stock data - so we replayed it



1 Morning

deliveries arrive - 20% of them late

2 Day

customers buy; if the shelf is empty the sale is lost

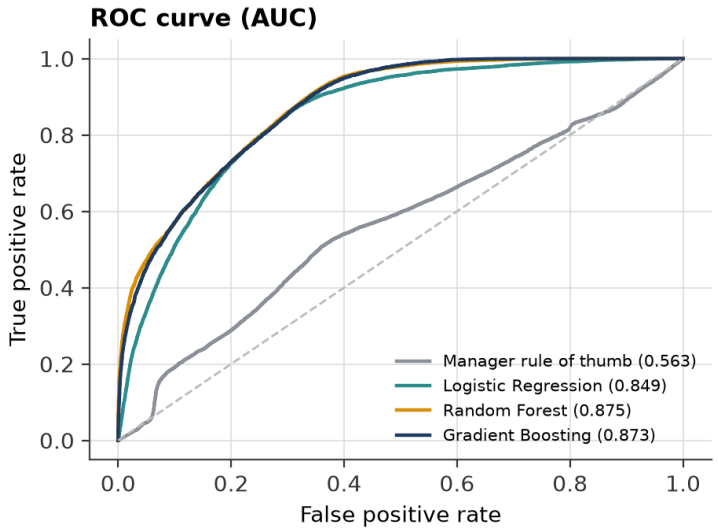
3 Evening

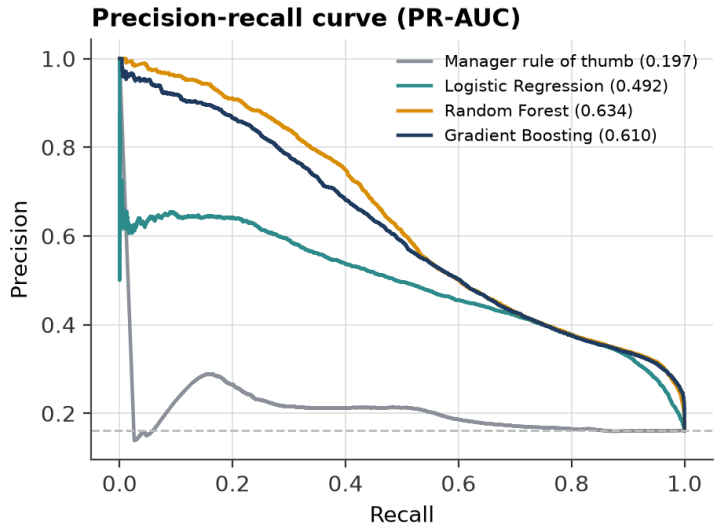
Mon & Thu: order up to a target level

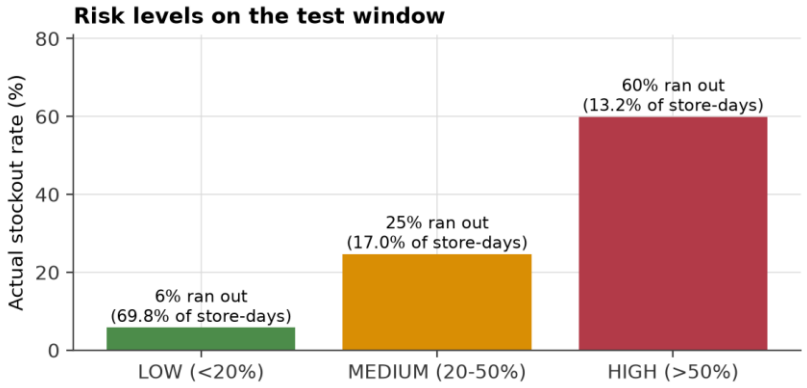
8.6%

stockout days under the manager's rule (last 4 weeks' average + 20%). We calibrated the buffer so the replay matches the 8.3% out-of-stock rate measured in real stores. Promotion days: 14% stockouts, other days 4%.

Spotting a stockout three days before it happens







HIGH stores really ran out 60% of the time; **LOW** only 6%.

0.873

ROC-AUC (rule of thumb: 0.56)

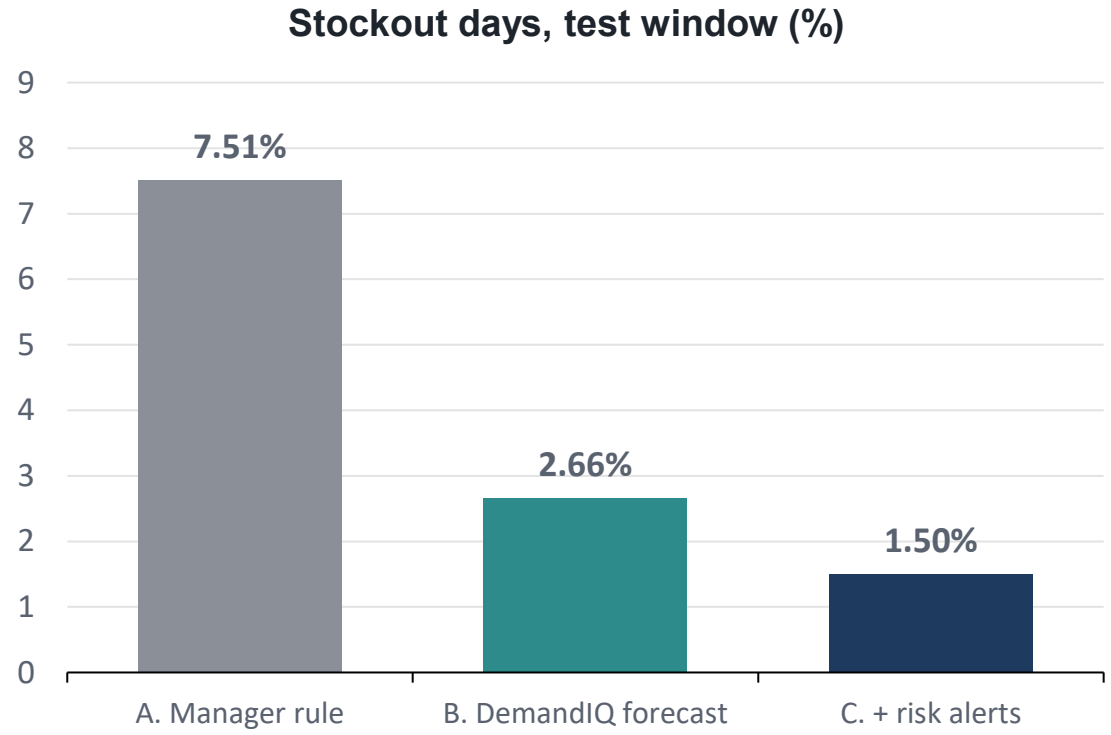
0.61

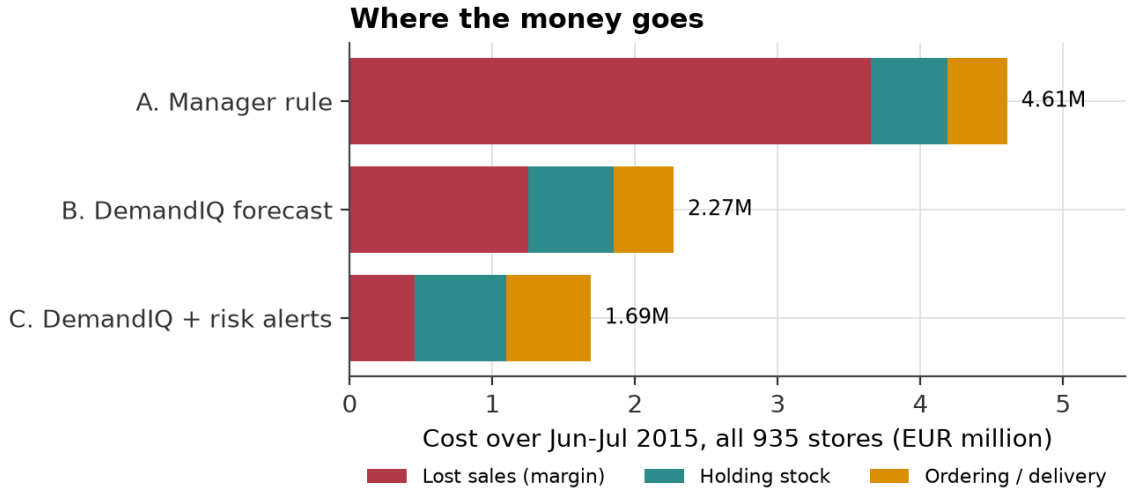
PR-AUC - over 3x the 16% base rate

60%

of stockouts caught at the tuned threshold

80% fewer stockout days, 63% lower cost

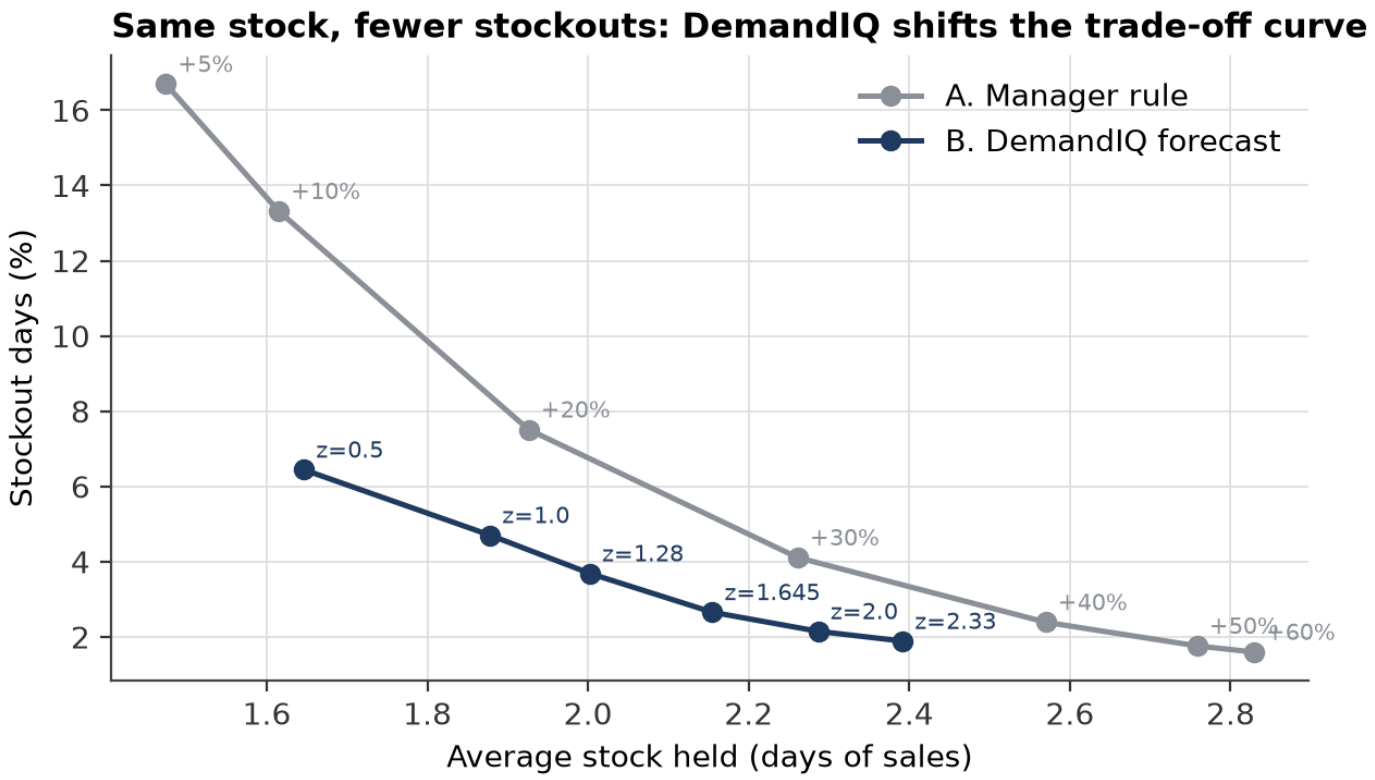




Policy	Fill rate	Stock (days)	Cost vs A
A	96.57%	1.93	+0%
B	98.82%	2.15	-51%
C	**99.58%**	**2.32**	**-63%**

A better forecast removes most stockouts; risk-triggered emergency orders remove much of the rest.

Same stock, fewer stockouts - it is not just "buy more"



The question

Could the manager do just as well by keeping a bigger buffer?

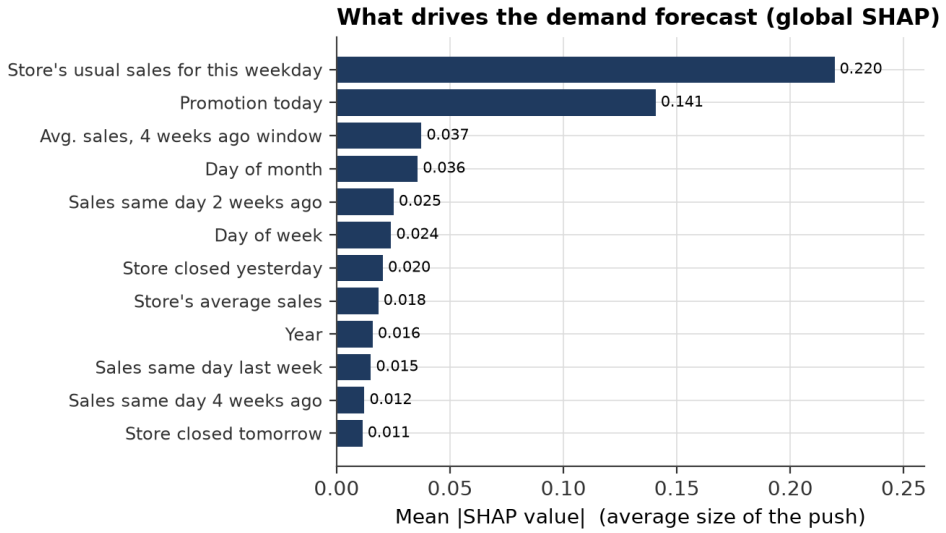
The test

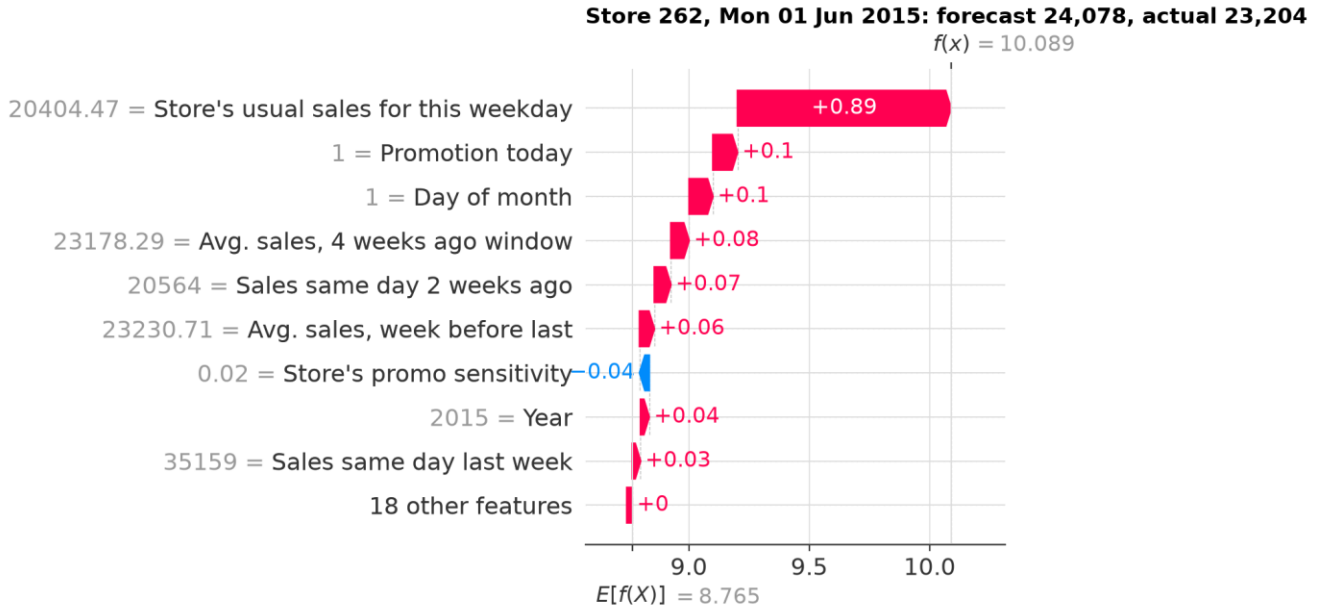
Replay both policies at many buffer and service-level settings.

The answer

At every level of stock, DemandIQ has fewer stockout days - it puts the stock on the right days, before promotions and holidays.

Every number comes with its reasons





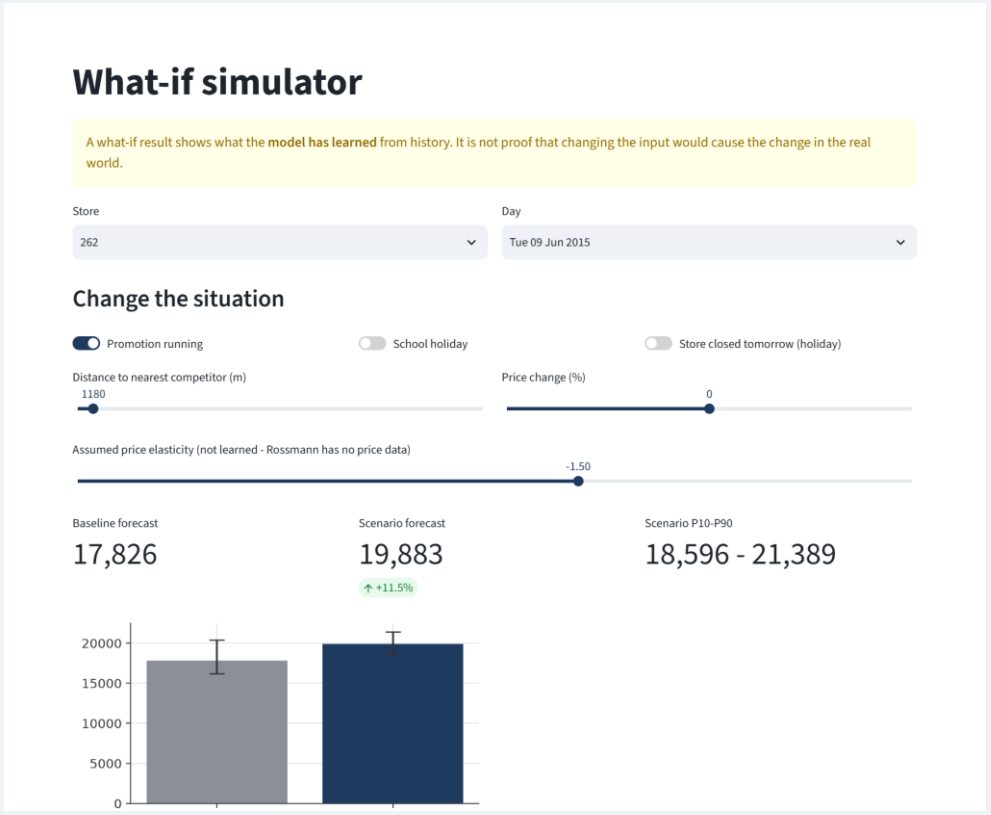
Global: what drives demand across all stores

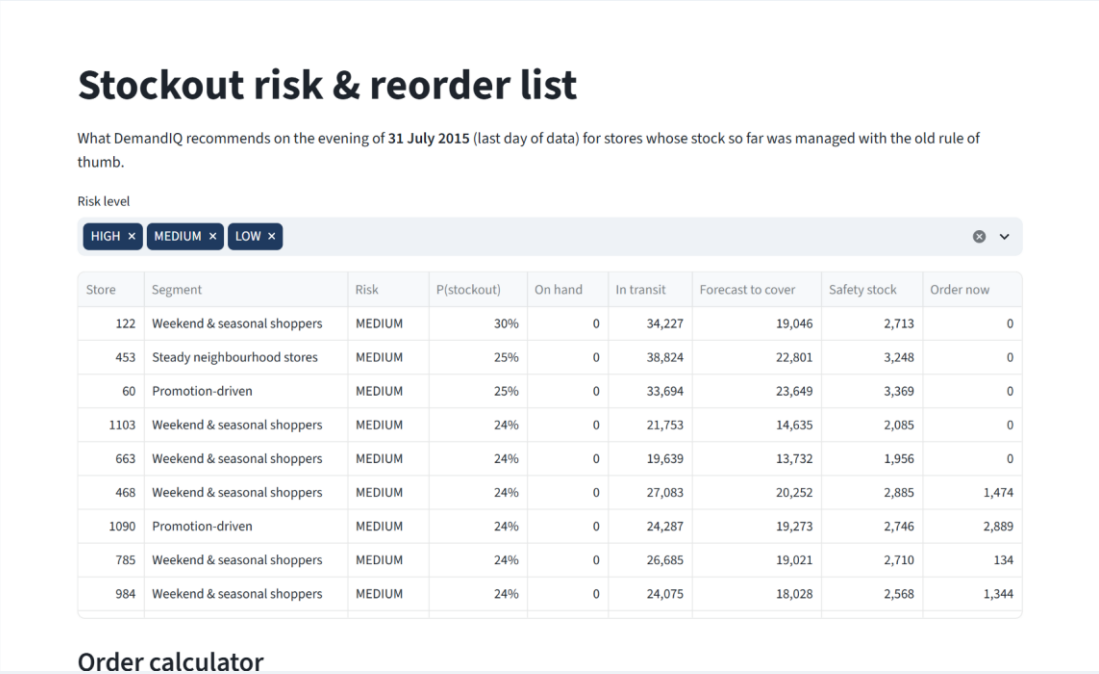
Local: why store 262 got 24,078 on Mon 1 Jun (actual 23,204)

What-if with the model: switching a promotion on lifts the forecast by 29% to 45% depending on the store segment. Shown as model behaviour, not proof of cause and effect.

DASHBOARD

A six-page Streamlit app that managers can use





Store	Segment	Risk	P(stockout)	On hand	In transit	Forecast to cover	Safety stock	Order now
122	Weekend & seasonal shoppers	MEDIUM	30%	0	34,227	19,046	2,713	0
453	Steady neighbourhood stores	MEDIUM	25%	0	38,824	22,801	3,248	0
60	Promotion-driven	MEDIUM	25%	0	33,694	23,649	3,369	0
1103	Weekend & seasonal shoppers	MEDIUM	24%	0	21,753	14,635	2,085	0
663	Weekend & seasonal shoppers	MEDIUM	24%	0	19,639	13,732	1,956	0
468	Weekend & seasonal shoppers	MEDIUM	24%	0	27,083	20,252	2,885	1,474
1090	Promotion-driven	MEDIUM	24%	0	24,287	19,273	2,746	2,889
785	Weekend & seasonal shoppers	MEDIUM	24%	0	26,685	19,021	2,710	134
984	Weekend & seasonal shoppers	MEDIUM	24%	0	24,075	18,028	2,568	1,344

Overview • Store forecast • Risk & reorder list • Why this forecast? • What-if simulator • Store segments

What we would do next

Limitations

1. Stock is simulated, not observed
2. Store level, not product (SKU) level
3. No price data - price lever uses a stated elasticity
4. P10-P90 band covers 73%, not 80%

Future scope

1. Validate on data with real stockout labels (FreshRetailNet-50K)
2. Indian retail with a festival calendar
3. Causal price effect with double/debiased ML
4. Weekly retraining and an evening HIGH-risk alert

CONCLUSION

A forecast creates value only when it becomes an explained decision

8.0%
demand forecast error (MAPE)

0.87
ROC-AUC for 3-day stockout risk

80%
fewer stockout days than the manager rule

63%
lower total inventory cost

Classical ML only • Random Forest, Gradient Boosting, Logistic Regression, K-Means • SHAP • Streamlit

Thank you - questions?

Yash Shah (L056) | Mahak Wadhawan (L074) | Yash Patil (L043) | Varshita Vaan (L069)